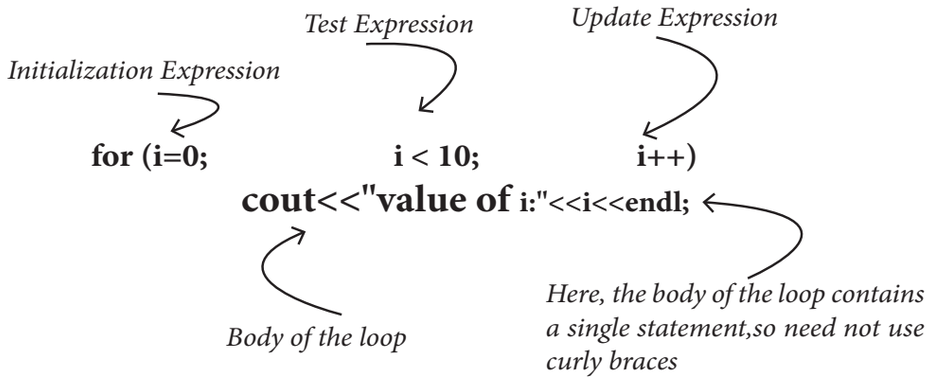
The following lines describes the working of the above given for loop:



In the above program, first the variable i is initialized, next i is compared with 10, if i is less than ten, the value of i is incremented. In this way, the numbers 0 to 9 are displayed. Once i becomes 10, it is no longer < 10. So, the control comes out of the for loop.

Illustration 10.8 C++ program to sum the numbers from 1 to 10 using for loop

```
#include <iostream>
using namespace std;
int main ()
{
int i,sum=0;
for(i=1; i<=10;i++)
 {
        sum=sum+i;
 }
cout<<"The sum of 1 to 10 is "<<sum;
return 0;
}
```

Output

The sum of 1 to 10 is 55

Variations of for loop

The **for** is one of the most important looping statement in C++ because it allows a several variations. These variations increase the flexibility and applicability of **for** loop. These variations will be discussed below:

Multiple initialization and multiple update expressions

Multiple statements can be used in the initialization and update expressions of **for** loop. These multiple initialization and multiple update expressions are separated by commas. For example,

```
#include<iostream>
using namespace std;
int main()
{
        int i, j;
            for(i=0, j=10 ;  i<j  ;  i++,j--)
            {
            cout<<"\nThe value of i is"<<i<<" The value of j is "<<j;
```

Multiple initialization expressions (separated by commas)

Multiple update expressions (separated by commas)

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